CSE275:OPTIMIZATION TECHNIQUES FOR MACHINE LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain the role of optimization in machine learning and artificial intelligence systems.

CO2 :: apply population-based optimization techniques to solve learning and tuning problems.

CO3 :: analyze evolutionary optimization methods for feature selection and model optimization.

CO4 :: apply swarm intelligence techniques to optimization problems in AI.

CO5 :: evaluate hybrid and metaheuristic optimization strategies for machine learning models.

CO6 :: develop optimization-based solutions for real-world machine learning problems.

Unit I

Introduction to Optimization in Machine Learning : Role of optimization in AI and ML, optimization problems in learning systems, loss minimization and search-based optimization, gradient-based vs gradient-free optimization, Convex vs non-convex optimization in ML, need for metaheuristic optimization, applications of optimization in feature selection, hyperparameter tuning and model selection

Unit II

Evolutionary Algorithms : fitness function, selection strategies, crossover and mutation operators, convergence behaviour, Premature convergence and diversity preservation, Fitness landscape intuition and search difficulty, applications of genetic algorithms in machine learning, feature selection using genetic algorithms, Introduction to evolutionary computation, genetic algorithms representation

Unit III

Swarm Intelligence Techniques : Swarm intelligence concepts, particle swarm optimization (PSO), ant colony optimization (ACO), artificial bee colony (ABC) and cuckoo search algorithm, exploration vs exploitation trade-off, Parameter sensitivity and tuning in swarm algorithms, Swarm stagnation and mitigation strategies, Parallel and distributed nature of swarm optimization, swarm-based optimization for machine learning problems

Unit IV

Advanced Nature-Inspired Optimization : Firefly algorithm and whale optimization algorithm, grey wolf optimization and grasshopper optimization algorithm, Conceptual grouping of metaheuristics comparison of metaheuristic algorithms, scalability and convergence issues in optimization

Unit V

Optimization for Machine Learning and AutoML : Hyperparameter optimization techniques, grid search vs random search limitations, evolutionary hyperparameter tuning, Bayesian optimization (conceptual), optimization for ensemble learning, introduction to automated machine learning (AutoML)

Unit VI

Hybrid Optimization and Applications : Hybrid optimization models, combining evolutionary and swarm-based approaches, real-world case studies in optimization-based ML systems, performance evaluation of optimization techniques, computational considerations in large-scale optimization

List of Practicals / Experiments:

List of Practical's

- To visualize and analyze convex and non-convex optimization landscapes using standard benchmark functions.
- To implement and compare gradient-based and gradient-free optimization methods for loss minimization in machine learning models.
- To implement a genetic algorithm for continuous function optimization and study its convergence behavior.
- To analyze the effect of genetic algorithm parameters on convergence and premature stagnation.
- To perform feature selection using genetic algorithms and evaluate its impact on model performance.
- To implement particle swarm optimization for solving continuous optimization problems.

- To compare swarm intelligence techniques (PSO, ACO, ABC) on standard benchmark optimization functions.
- To apply swarm-based optimization techniques for hyperparameter tuning of a machine learning model.
- To implement and evaluate advanced nature-inspired optimization algorithms such as Grey Wolf or Whale Optimization.
- To compare evolutionary hyperparameter optimization with grid search and random search techniques.
- To implement Bayesian optimization for efficient hyperparameter tuning of machine learning models.
- To design and evaluate a hybrid optimization-based machine learning solution on a real-world dataset.

Text Books:

1. PRINCIPLES OF SOFT COMPUTING, 3ED by S. N. SIVANANDAM, S. N. DEEPA, WILEY

References:

1. ESSENTIALS OF METAHEURISTICS by SEAN LUKE, LULU.COM